

at, so we don't get subjected to some shops and establishments or factories building code that don't necessarily apply."

And then there's water, the quiet gating item in every fab story. "What do you do with the water? You can't pump it down a river. You can't pump it down to the sea," says Avi, and rightly so. His benchmark is almost aggressively practical: "Is there a parallel policy framework to say, okay, if you want to build this facility in this location, here are the 10 things we will get done within a month. Can we do that? I think that speed is needed."

Forming extremely safe closed loop systems is the need of the hour, if India wants to be serious about setting up fabs. Which is where knowledge transfer and policy recommendations from companies like Applied Materials that have been there and done are crucial for India's semiconductor manufacturing ambitions. Speed of execution is a key component of this puzzle, according to Avula.

"Singapore did it over the last 20-25 years. China did it in 20 years. I don't think India has that kind of time. There are challenges in export of technology, and certain countries are already restricted. Slowly, those barriers and those walls may get higher. So, you don't have time," cautions Avula.

Avula doesn't buy the panic around Indian talent. "Actually, I'm not too worried about the talent part. I think we have a fantastic educational system." He refuses to let "skilling" become a distraction from harder system-building: "I think we can train them on the job. We do that all the time."

He closes by insisting on resolve, not despair. "I don't feel gloomy. People in the government at multiple levels get the seriousness and are on it. I think this government recognizes that not being a player here means you're always going to be on the receiving end of somebody else's technology. So if you don't develop it, you're just dependent. Who knows, one day that fighter jet you bought from some other countries is just going to get switched out remotely. What are you going to do with that?"

For Avula, that fighter-jet image is the point of the entire exercise. Chips aren't just another industrial trophy; they're the plumbing of modern power. If you don't control some of the plumbing, you're permanently negotiating from the receiving end – on prices, on access, on timelines, and on what you're allowed to build. Which is why Avula stresses India needs strategic silicon with a real, running ecosystem – built fast, built safely, and built in a way that makes the world keep showing up. **d**

APPLIED MATERIALS: KEY HIGHLIGHTS

OVER 20+ YEARS IN INDIA'S SEMICONDUCTOR ECOSYSTEM

Established in 2002, Applied Materials has built over two decades of sustained presence in India's semiconductor and manufacturing landscape.

STRONG NATIONAL FOOTPRINT ACROSS 5 ESTABLISHED REGIONS

Operations span Bengaluru, Chennai, Mumbai, Coimbatore and Pune, with virtually every global function supported from India.

\$400M INVESTMENT DRIVING ADVANCED R&D AND AI IN INDIA

A \$400M investment commit positions India as a global hub for next-generation semi R&D, accelerating innovation, commercialization & full product development at scale.

COMPREHENSIVE PRODUCT DEVELOPMENT INFRASTRUCTURE

Facilities include exploration and prototype labs, reliability and validation testing, software development and qualification, and a Class 1K Process Development Center clean-room for high-precision work.

UNIQUE 300MM WAFER PROCESSING & AI CAPABILITIES

India Validation Center enables full 300mm wafer processing for design and testing, while the AI CoE supports high-volume manufacturing through AI, ML and advanced analytics.

APPLIED VENTURES EXPANDS STRATEGIC FOOTPRINT IN INDIA

The venture capital arm of Applied Materials has made three strategic investments in deep-tech startups and funds in India. It has also joined India Deep Tech Alliance as a Strategic Partner.